

Understanding travel behavior and mode choice prediction for university commuters: insights from discrete choice models and machine learning

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ABSTRACT

While there has been a growing interest in studying the commuting behavior of university students using mode choice models, most existing studies still lack the in-depth analysis needed to fully understand the factors influencing their transportation mode choices. To address this gap, this study examines the effects of various built-environment factors, socio-economic factors, and travel information to explore the commuting behavior of university commuters. Various prediction models based on discrete choice models (DCMs) and machine learning (ML) techniques are developed to predict the travel mode choice of university commuters. The developed prediction models are tested using a Commuting Habits survey conducted at a Canadian university. The results indicate that for transit and walk, household attributes such as car ownership, household size, and income level have significant impacts on choosing these modes as primary modes of transportation. It was also observed that car ownership and availability of transit passes are the most influential factors. As such, policymakers can further examine these relationships to incentivize commuters to promote sustainable transportation modes. Comparing DCM and ML-based models to predict the mode choice of university commuters, our analysis revealed that although the multinomial logit discrete choice (MNL) model resulted in lower performance in predicting commuters' travel mode choices, it provides more insightful information regarding each specific mode of transportation. ML models prioritize information differently when making predictions, with MNL placing more emphasis on socio-demographic variables and household characteristics of respondents, whereas ML algorithms rely more on revealed travel-related variables.

Introduction

University campuses, similar to other densely populated areas, exhibit a high concentration of human activities, making them significant sources of trip generation (Shannon et al., 2006). The large volume of journeys to and from campuses leads to various environmental impacts on the university and surrounding areas, as noted by Tolley (1996) and Shaw et al. (2013). For instance, the considerable use of personal cars contributes to the high occupancy of on-campus parking lots (Rosenfield, 2018), causing car commuters to resort to on-street parking lots outside university property. This, in turn, affects neighboring communities with traffic congestion, pollution, parking difficulties for residents, and other issues. Furthermore, traffic problems such as traffic jams during peak hours, noise, pollution, and greenhouse gas emissions are among the other undesirable consequences of personal car use (Marsden and Rye, 2010; Ory et al., 2004; Turcotte, 2005; Wallace et al., 2011).

Moreover, research has shown that university commuters exhibit different mobility behaviors compared to the general population. A study by Balsas (2003) highlighted the importance of university commuting by comparing eight commuting surveys conducted at various universities in the US. The study concluded that understanding university commuters' travel behavior is crucial, as it differs significantly from that of the general population. In line with this, Khattak et al. (2011) confirmed the finding from a previous study, emphasizing that the travel behavior of university commuters is distinct from the general population by examining commuting data from four major universities in Virginia, US. The study also pointed out that the travel behavior of university commuters is neither well understood nor adequately represented in travel demand analyses.

Therefore, it is imperative to examine university commuters' behavior in order to impact campus communities positively, promote more sustainable transportation systems, and enhance urban sustainability. Travel mode choice modeling is a crucial topic in travel behavior

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studies as it offers insights into both commuters' mode choice preferences and the factors affecting their decisions, significantly aiding the transportation planning process. Travel mode choice is also considered a vital step in the demand forecasting process, which is required for policy making.

The objective of mode choice prediction is to forecast individuals' travel mode preferences by considering various factors. In this study, we developed several machine learning (ML) techniques, including decision tree (DT), extreme gradient boosting (XGBoost), random forest (RF), and artificial neural network (ANN) to predict transportation mode choice using data obtained from the university commuters in 2020. Additionally, we developed a multinomial logit discrete choice model (MNL), which is traditionally employed in travel demand modeling, and compared its performance with advanced ML algorithms. Apart from predicting transportation mode choice, we examined a variety of influential factors, categorized into individual levels, household levels, built environment-related, and travel-related, on university commuters' decisions to provide deeper insights into their attitudes.

In recent years, ML methods have been used more often in travel mode choice studies because they can handle complex relationships in the data. Unlike traditional logit models, ML methods can capture non-linear effects without needing a fixed utility structure (Martín-Baos et al., 2023; Zhang et al., 2023). Some studies have shown that tree-based methods such as random forests and gradient boosting predict travel modes more accurately than MNL, although MNL is still easier to interpret (Martín-Baos et al., 2023). Other work has also found that ML methods can better handle high-dimensional datasets and interactions among socio-demographic, built environment, and travel-related variables, which are often difficult to represent in a traditional logit model (Xie et al., 2022). We have even seen ML techniques being used for university commuters mode choice such as Díaz-Ramírez et al., 2023. They proposed a two-phase methodology combining a descriptive analysis of mode choice preferences with a predictive phase using decision tree-based classifiers. These strengths are useful in the study of university commuting, where many different personal, household, and travel factors influence decisions (Akter and Alam, 2024; Xie et al., 2022; Zhu et al., 2023). In this study, we compare ML methods with the MNL model to improve prediction accuracy and to gain more insight into the factors that shape commuting choices.

However, there is a research gap in understanding the trade-offs between traditional discrete choice models and modern machine learning methods in this domain. Prior studies indicate that machine learning can improve prediction accuracy in travel mode choice (Martín-Baos et al., 2023) and capture complex non-linear relationships that a logit model might miss (Xie et al., 2022). At the same time, multinomial logit models remain valuable for their interpretability and mode-specific insights. Few studies have directly compared an MNL with various ML algorithms on the same problem to determine not only which approach predicts better, but also how their results differ in terms of explaining travel behavior. This comparison is particularly relevant for university commuters, whose behavior can diverge from the general population (Díaz-Ramírez et al., 2023). In this study, we address that gap by systematically evaluating both an MNL model and several ML models on a university commuting dataset. This allows us to assess the extent of any predictive improvements offered by ML and to identify whether the ML techniques uncover additional or different determinants of mode choice. By doing so, we seek to demonstrate how data-driven ML methods might complement traditional behavioral models, providing a more complete understanding of university travel mode preferences.

Literature review

To effectively establish the foundation of this research, it is crucial to explore two dimensions within the literature. First, it is necessary to examine the factors affecting university commuters' travel behavior, as their behavior is significantly different from the general population.

Second, it is essential to review the methodologies and modeling techniques employed for predicting individuals' mode choices. This will help determine the models that best fit the dataset and yield superior performance.

Influential factors on university commuters' travel behaviour

A wide variety of factors have been found to influence university commuters' travel behavior in the literature. Schneider and Willman (2019) reported that socio-demographic characteristics affect university commuters' attitudes toward different transportation modes. Among these characteristics, age and gender have received the most attention. Nguyen-Phuoc et al. (2018) underscored the importance of age and gender on transportation mode choice decisions among students from six universities in Danang, Vietnam. The study found that older students and male students are more likely to use active transportation modes. Similarly, Zhou (2012) and Eluru et al. (2012) demonstrated that gender is significantly associated with active mode and public transit usage among university students in Los Angeles and at McGill University, Canada, respectively. Other influential socio-demographic factors discussed in the literature include household income (Hamad et al., 2021; Nguyen-Phuoc et al., 2018), household car ownership (Eluru et al., 2012), and university status (Davison et al., 2015; Zhou, 2012). Regarding university status, Eom et al. (2009) concluded that the students' level of study significantly affects the travel demand model for North Carolina State University students. Moreover, Davison et al. (2015) showed that part-time students are more inclined to use active transportation modes.

In addition to socio-demographic factors, a variety of other factors have been found to influence university commuters' travel behavior. Built-environment factors, such as travel distance, availability of public transit facilities, and active mode infrastructures, have proven to be influential in university commuters' mode choice (Davison et al., 2015; Rérat, 2021). Concerning travel distance, Wang and Liu (2015) identified the factors affecting the University of Queensland commuters' mode choice and concluded that travel time and distance are the most influential factors in their decision-making process. Zhou (2012) reported that commuting distance is positively correlated with carpooling and telecommuting among a sample of students from Los Angeles universities. Similarly, Eluru et al. (2012) indicated that factors affecting travel time, such as travel distance, the number of transfers, and walking time, significantly influenced university commuters' decision to choose public transit. Infectious-disease risk can also be embedded in generalized cost via crowding penalties; in our setting, longer trip distance proxy higher exposure duration, helping explain the sensitivity of transit choice (Devasurendra et al., 2022). Shannon et al. (2006) also found that university commuters' travel patterns could be significantly affected by their perceived travel time.

With respect to the availability of public transit and active mode infrastructure, Davison et al. (2015) demonstrated that the availability of public transportation facilities can significantly affect students' travel behavior. Rérat (2021) indicated that improvements in public transport infrastructure at the University of Lausanne led to a significant reduction in car use among the university community. Shannon et al. (2006) analyzed a one-week travel survey at the University of Western Australia and discovered that policy applications, such as subsidized public transport passes, parking costs, and bicycle network infrastructure, are the most critical factors influencing commuters' transportation mode choices to and from the university. Zhou (2012) concluded that owning a transit pass and being multimodal would increase the likelihood of using alternative modes while possessing parking passes would increase the probability of using a car.

Other factors identified in the literature as impacting university commuters' mode choice include housing rents on or near campus (Shannon et al., 2006), external events such as the COVID-19 pandemic (DeWeese et al., 2022; Lodi et al., 2024; Soltani et al., 2024), and the

value of travel time (Basu and Hunt, 2014). Furthermore, Hossain et al. (2022) investigated the commuting behavior of post-secondary students in the Greater Toronto and Hamilton areas, discussing that students' daily travel habits are directly influenced by how frequently they use different transportation modes.

While these studies provide valuable insights, they often lack a comprehensive examination of the combined effects of socio-demographic, infrastructural, and travel-related factors on mode choice. Additionally, there are various factors whose impacts on university commuters' travel behavior have not yet been extensively studied. To address this gap, this study examines the effects of various built-environment factors (e.g., weather conditions, number and density of bus stops in proximity to the origin, and number and density of metro stations in proximity to the origin), socioeconomic and demographic factors (e.g., household size, accessibility to cars, accessibility to bikes, and number of cars owned), and travel information (e.g., weekly travel frequency, distance, and duration to campus using car, transit, bike, and walking). Additionally, the influence of age, university status, and educational level on university commuters' travel behavior is further investigated, as these factors have not been widely examined in the literature.

Transportation mode choice prediction approach

As previously discussed, predicting travel mode choice is a crucial aspect of travel behavior prediction studies, as it offers insight into both commuters' mode choice preferences and the factors influencing their decisions. Researchers have predominantly employed two groups of modeling techniques to analyze travelers' mode choice behavior: 1) discrete choice modeling and 2) ML techniques. This section reviews the prior research conducted in transportation mode choice modeling.

Traditionally, mode choice studies have relied on discrete choice models, particularly logit models. Logit-based models are grounded in random utility theory, where the likelihood of individuals choosing a specific alternative is based on their socio-economic attributes and the relative attractiveness of the alternative (McFadden, 1981). By employing the concept of utility, we can gauge the attractiveness of alternatives and represent it as a numerical measure of a choice's appeal to an individual. The MNL model is widely recognized as the most common DCM employed in the context of transportation mode choice modeling. It is a single-level logit model with error terms assumed to be independent and identically distributed (Ben-Akiva et al., 1985). The MNL model has been extensively utilized in the literature for modeling travel mode choice. Train (1980) concentrated on joint modeling of car ownership and transportation mode for commuting to work. They discovered that households closer to their workplaces or with greater access to public transportation have a lower car ownership rate. Anas (1981) provided an estimation of joint location and travel mode choice using the MNL model with aggregated data. These two study models are considered early applications of transportation mode choice modeling using the MNL model. In another study, Koppelman (1983) developed the MNL model to predict travel mode share in response to transit service changes and the introduction of a new transportation mode.

Given the various sources of uncertainty in the analysis of travel mode choices, it is clear that the limitations of Multinomial Logit (MNL) and Machine Learning (ML) models must be considered. The applicability of a single model may not be the optimal approach for predicting travel mode choices. This perspective aligns with the findings of Aboutaleb et al. (2021), who pointed out the intrinsic limitations of supervised machine learning models in policy analysis contexts where extrapolation of multiple options is essential. This is further complicated by the differences in size and structure across various datasets, which corroborates the notion that a single model may not always be the most effective strategy for travel mode prediction (Zeng et al., 2018; Wang et al., 2021). As such, it may be more insightful to develop and utilize multiple predictive ML and DCM models and select the one that

performs best in a given context.

Salas et al. (2022) noted the increasing use of machine learning methods in the transportation sector. In most comparative studies, ML algorithms have been demonstrated to be more promising and accurate than DCM (Aghaabbasi et al., 2023; Cheng et al., 2019; Hagenauer and Helbich, 2017; Ma and Zhang, 2020; Salas et al., 2022; Xia et al., 2023; Zhao et al., 2020). Few studies in the literature indicate that DCM outperforms ML models in prediction accuracy (Hensher and Ton, 2000). To the best of our knowledge, there is no definitive evidence that either DCM or ML models consistently outperform the others. However, there are some benchmarks for researchers to rely on for mode choice prediction analysis. Pineda-Jaramillo (2019) reviewed more than 100 papers that developed predictive models on transportation mode choice and highlighted that most authors rated the MNL model as having the lowest accuracy among predictive modes. The paper specifically introduced RF algorithms as the most promising method for predicting travel mode choice among ML and the MNL model. In another study, Wang et al. (2021) evaluated more than 100 classifiers from 12 ML and DCM families on three datasets. In terms of prediction accuracy, neural networks, ensemble methods (i.e., RF, boosting, and bagging), and DT ranked first to fifth among 12 model families, and discrete choice models ranked seventh. In terms of computational time, a considerable difference between DCM and ML algorithms was observed. The study showed that DCM models are 30 times slower than the slowest ML model on average; thus, more attention needs to be paid to enhancing the computational efficiency of DCM. The study also concluded that the relative rankings between ML and DCM classifiers show great stability, meaning that the order of algorithms does not vary while testing on different datasets.

Several studies have been performed on the transportation mode choice prediction of the general population and compared the performance of various algorithms for predicting individuals' transportation modes. In contrast, studies investigating university commuters are primarily limited to descriptive analysis and did not develop prediction models to deeply investigate the factors affecting university commuters' transportation mode choice decisions. As Balsas (2003) also concluded, university commuters' behavior differs significantly from the general population. Thus, it is necessary to explore the transportation mode choice as well as other aspects of university commuters' behavior.

Survey data and methods

This section discusses the various data and model-level steps conducted in this research's methodology. It presents the steps considered to conduct the survey and prepare a dataset that can accurately represent commuters' behavior. Additionally, it explains the theory of the MNL model, the most commonly used discrete choice modeling technique in the literature, and four selected ML classification algorithms developed in this paper.

Case study and data collection

The university being studied in this paper is a public research institution located in Alberta, Canada. More than 33,000 students and 7,000 employee members visit the university campuses every day (University of Calgary, 2019). The university has four campuses in the city: Main Campus, Foothills Campus, Spy Hill Campus, Downtown Campus, and one Campus in Doha, Qatar. This study focuses on the four campuses located in the city (Fig. 1).

It is essential to analyze the commuter behavior of the university commuters for three main reasons. Firstly, aside from being the second-largest university in Alberta, the University has the largest community in the city. Secondly, there has never been a detailed analysis of the commuting behavior of the university commuters. Thirdly, since the University has four campuses in the city, the study can analyze how different urban structures and transportation services affect commuters

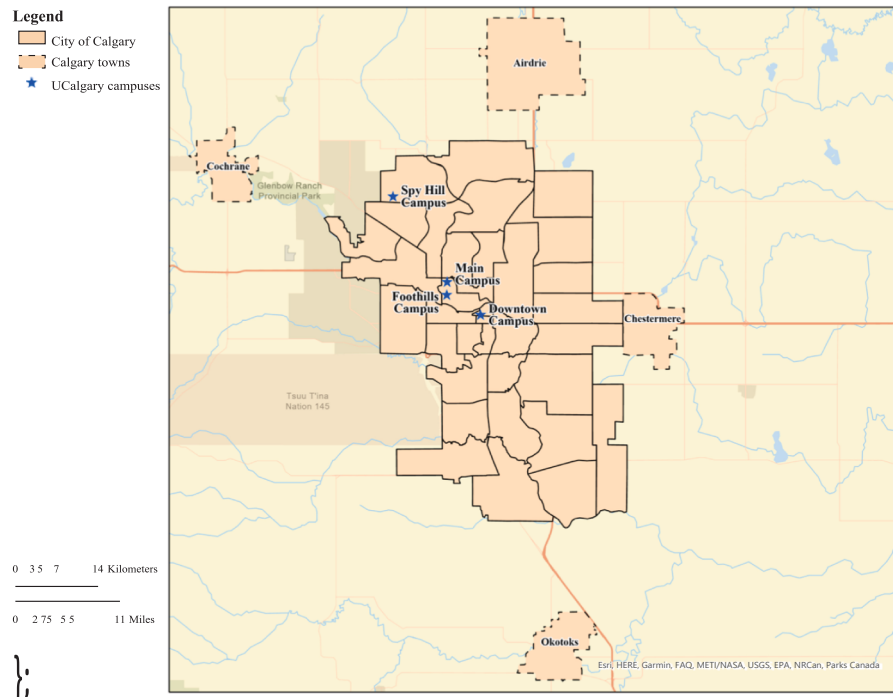


Fig. 1. Postal code boundaries and the university campuses locations.

to the four campuses. Thus, we collected the required data through the 2020 commuting habits survey.

Between Winter 2020 and Fall 2020, the University commuting habits survey was conducted to evaluate the commuting behavior of respondents 18 years of age and older. The University Conjoint Faculties Research Ethics Board approved this survey to be done online. The survey was hosted online using QualtricsXM and was distributed by email and social media. A total of 979 respondents participated in the survey, with 867 records considered for the modeling after pre-processing. This data consists of 430 students, 368 staff members, and 69 faculty members.

Survey design

The online survey consisted of 45 questions divided into five different sections. The first section focused on gathering spatially related information, such as the respondents' home postal code, type of residency, and the campus where they spend the majority of their time. This information is necessary for modeling the respondents' travel mode choices, as it, directly and indirectly, influences their decisions. In this section, respondents who use transit or private cars to travel to the Main Campus were asked to indicate their final transit stop/station or parking lot on the Main Campus map, respectively. This information helps the university identify popular transit stops/stations or parking lots for various commuters and provides insight into which group of commuters prefer to park their cars in street lots and which spots are more appealing to commuters.

The second section of the survey aimed to gather information about respondents' commuting patterns. Respondents were asked detailed questions about their primary means of transportation between home and campus, as well as between different campuses. This included questions about the accessibility of private cars and bicycles, the frequency of their commute to the university during the week, and the transportation mode they mostly use. This section provided a comprehensive understanding of the university commuters' behavior and the required information for transportation mode choice modeling. The most common transportation mode identified by respondents would be considered as labels in predictive modeling.

After the commuting pattern section, respondents were asked to rank the top factors/barriers preventing them from using each transportation mode. These barriers were chosen based on the nature of each transportation mode and the literature. This section captured the most significant factors preventing respondents from using modes they did not use. Additionally, this section included questions to gather respondents' attitudes toward various university transportation programs, such as bike/scooter sharing and transit UPass (transit pass of the University).

The fourth section of the survey focused on commuters' level of satisfaction with different transportation modes. The survey provided specific attributes for each transportation mode based on its nature and asked respondents to rate their satisfaction on a 5-point Likert scale, from strongly satisfied to strongly dissatisfied. Unlike the previous section, satisfaction-related questions were only asked from respondents who declared that they use that specific transportation mode as their primary means of travel.

Finally, the survey collected socio-economic data from respondents, including age, income, gender, university status, etc. This data was used to group respondents and compare travel behavior for different groups.

Survey feature selection

Developing predictive models involves selecting explanatory features from the numerous variables available in the dataset. Identifying the appropriate number of input features from all available variables reduces the computational cost of modeling and enhances model performance. Consequently, it is crucial to select only the features that describe the model's output, in this case, the transportation mode. Explanatory variables can be attributes that influence individuals' decisions to choose a transportation mode for their travel. As discussed in the literature review section, travel mode choice is affected by various factors, including individual and household characteristics, built environment, trip attributes, and mode-specific attributes.

The University 2020 commuting survey gathered comprehensive information concerning the individual and house-hold characteristics of university commuters. With respect to trip attributes and built-environment factors, zonal data of respondents' living locations were extracted from the survey. This data represented an area of the city

encompassing several neighborhoods without overlapping each other. The zonal data was utilized to generate new features that could describe travel information attributes and built-environment factors. In this process, we considered 25 variables (Table 1) related to respondents' built environment and travel conditions, which were either suggested by the literature or displayed an effect in the descriptive analysis.

a) Built-environment attributes

Ten latitude/longitude points were randomly generated within each postal code area, and the Google Maps Distance Matrix API was employed to estimate travel time by car, transit, bike, and walk from each generated location to all four university campuses during the AM peak. The average travel time from these ten points to the university campuses was considered the estimated travel times of each postal code area to the university. Ultimately, the estimated travel time of the primary transportation mode of each respondent and three other unchosen modes were assigned to each respondent based on their living postal code and the campus they primarily commute to.

In addition to travel time duration, the same method calculated the average travel distance from each postal code area to different campuses. After determining travel time-related variables, they were assigned to each respondent based on the first three digits of their living location postal code and the campus they mainly commute to.

b) Travel information attributes

Two datasets provided by city transit agency were used to count the number of LRT stations and bus stops in each postal code area. The datasets offered the geographical latitude and longitude of each transit stop/station in the city. These datasets were joined with the city's postal code boundaries layer using the "spatial join" tool in ArcGIS Pro software (version 2.6.0) to count the number of transit stops/stations within each boundary. Based on the number of transit stops/stations in each postal code and the postal code area, the density of bus stops and LRT stations in each postal code area was calculated. The bus stop and LRT station density were assigned to respondents as new features based on their living location. For training the models, the postal code feature was replaced with the generated features.

Categorical features

After selecting variables, it is necessary to handle categorical data. To build travel mode prediction models, it is crucial to distinguish between numerical and categorical variables. Numerical variables refer to real numbers that have continuous impacts on the learning target, while categorical variables consist of fixed values representing categories. In order to use predictive classification algorithms, categorical variables should be converted into numerical forms. This process is typically referred to as encoding. The two most commonly used methods for encoding categorical data are one-hot encoding and label encoding.

Both techniques have specific advantages and disadvantages, and there is no consensus in the literature favoring one over the other (Ma and Zhang, 2020). One-hot encoding is not appropriate for features with many categories, as it increases the dataset's sparsity, and different categories of the variables lose their relationship in model training. On the other hand, label encoding implies a mathematical order to the categorical variables, which is inappropriate and can cause misunderstandings during training. In this paper, categorical features that have a mathematical order in their nature (e.g., study level) are encoded using the label-encoding approach, while the rest are encoded using the one-hot encoding approach to avoid sparsity and prevent implying incorrect relationships in the dataset.

Applied models

In recent years, there has been a growing interest in exploring machine learning-based classifications for transportation mode choice modeling as an alternative to discrete choice modeling. This shift is attributed to advancements in technology, leading to a substantial increase in the availability of transportation-related data. The presence of

Table 1

Exploratory variables for travel mode prediction.

Variable group	Coding	Description
Individual attributes	Uni-Status	University status (student, staff, or faculty member)
	Age	Age of survey respondents (9 ranges)
	Gender	Female or male
	Edu-Level	Students' educational level (bachelor, master, or Ph.D.)
Household attributes	Transit-Pass	Holding transit pass
	Size	Number of individuals living in a household
	Income	Annual household income (10 ranges)
	Car-Acc	Accessibility to car for commuting
Built-environment	Bike-Acc	Accessing bike for commuting
	Car-Own	Number of cars owned by the household
	Campus	Primary commuting campus (4 available options)
	Season-Effect	Whether season affects commuters or not
Travel information	Bus-Stop	Number of bus stops in the vicinity of living location
	Bus-Density	Density of bus stops in the vicinity of living location
	LRT-Stop	Number of LRT stations in the vicinity of living locations
	LRT-Density	Density of LRT stations in the vicinity of living locations
	Travel-Freq	Weekly travel frequency to campus
	Drive-Dist	Driving distance to the primary campus
	Drive-Dur	Driving duration to the primary campus
	Transit-Dist	Transit distance to the primary campus
	Transit-Dur	Transit duration to the primary campus
	Walk-Dist	Walking distance to the primary campus
	Walk-Dur	Walking duration to the primary campus
	Bike-Dist	Biking distance to the primary campus
	Bike-Dur	Biking duration to the primary campus

data facilitates a novel approach to predicting mode choice, where ML classifiers can learn the intricate relationships between data points and maximize the model's prediction accuracy. The literature suggests a variety of ML algorithms for predicting travelers' mode choice, including K-nearest neighbor (Jahangiri and Rakha, 2015), Support Vector Machine (Xian-Yu, 2011), DT family models (i.e., DT, RF, and XGBoost) (Tang et al., 2015; Cheng et al., 2019; Wang and Ross, 2018), and ANN (Ma and Zhang, 2020).

The following sections offer a brief overview of the MNL model as the baseline for this research, and the four classes of supervised classification algorithms employed in this paper: DT, RF, XGBoost, and ANN.

Multinomial logit model

The baseline of our study is the MNL model. To ensure proper model identification, all socio-demographic and person-specific variables were specified as alternative-specific, meaning they were interacted with mode dummy variables so that each mode has its own coefficient. For example, 'Travel-Freq' has different effects for transit, walk, car, and bike. In addition, we fixed the constant of the car mode to zero, which is a standard normalization in multinomial logit models. As explained by Domencich and McFadden (1975), the utility function in these models is only identified up to an arbitrary additive constant. By setting one mode's constant (here, car) to zero, we establish a reference point, which prevents redundancy and ensures that the utilities of the other modes are interpreted relative to this baseline. This normalization allows the model to be estimated correctly and provides a clear frame of comparison across transportation modes.

Developing an MNL model with the highest prediction accuracy requires careful construction of the observed component of the utility function, which depends on selecting the optimal combination of available features for each transportation mode. This process involves iterative testing and relies on engineering judgment. In this study, 47

distinct combinations of features were evaluated, and the observed utility function that produced the highest prediction accuracy is presented in the equation below:

$$V_j = \beta_{cj} + \beta_{1j}\text{Travel-Dist} + \beta_{2j}\text{Travel-Freq} + \beta_{3j}\text{Season-Effect} \\ + \beta_{4j}\text{Transit-Pass} + \beta_{5j}\text{Edu-Lvl} + \beta_{6j}\text{Gender} + \beta_{7j}\text{Age} + \beta_{8j}\text{Car-Own} \\ + \beta_{9j}\text{Size} + \beta_{10j}\text{Income} + (\beta_{11}\text{Bus-Density} + \beta_{12}\text{LRT-Density}) \quad (1)$$

Where:

V_j = the observed utility for transportation mode j .

β_{cj} = the alternative specific constant that sets the default bias towards a particular alternative j .

β_{ij} ($i = [1, 12]$) = the coefficient of variables for specific alternative j .

On top of these variables, the MNL model used the Car-Acc and Bike-Acc variables as determinant variables for the car and bike utility functions. It means that the observed utility functions of car and bike were coded in a way that they become zero in case the commuter has no access to these modes.

Although University Status (student vs. staff/faculty) was recorded in the survey, it was excluded from the final MNL model. In our experiments, adding this variable did not yield a significant effect on mode choice after accounting for other variables, nor did it improve model fit. For model simplicity and to avoid multicollinearity, we removed Uni-Status from the MNL specification, its influence appears to be largely captured by age, income, and other included factors.

Travel time was indirectly incorporated via travel distance metrics. Due to multicollinearity between time and distance, the model used distance as a proxy for travel time, an approach supported by the significance of distance in our results. Unfortunately, travel cost could not be included explicitly, as the survey did not collect cost data (e.g., fares, fuel expenses). We relied on proxies such as transit pass availability and car ownership to reflect cost and convenience factors. We acknowledge that excluding explicit cost is a limitation; however, many students benefit from a prepaid UPass and parking costs were only indirectly reflected (through reported parking concerns). Future work should consider collecting and including travel cost data, as prior studies have shown subsidies, transit fare discounts, and parking fees to be critical policy levers influencing mode choice (Zahabi et al., 2012; Litman, 2004).

Decision tree

The DT classifier is based on the classification and regression trees (CART) algorithm introduced by Breiman et al. (1984). This algorithm employs recursive binary splits, with each split chosen to minimize the randomness of the data at that specific point. In this study, the full pure DT model is developed using the Gini Impurity index (Atkinson et al., 1970) on the training dataset utilized for training the models. The mathematical formulation of the Gini impurity index (i.e., G) is presented below:

$$G = \sum_{j=1}^C P_j(1 - P_j) \quad (2)$$

Where:

C = total number of labels

P_j = is the probability of picking a data point with label j

Building trees with all pure nodes does not necessarily guarantee the best performance on the test dataset due to overfitting. In this study, the pruning technique is employed to prevent overfitting of the model to the training dataset. After pruning the tree, stratified 10-fold cross-validation is utilized to find the most robust and validated model, as recommended by Kohavi et al. (1995).

Extreme gradient boosting & random forest

Both XGBoost (Friedman, 2001) and RF (Breiman, 2001) are tree-based ensemble methods that can be employed for classification.

Ensemble learning methods combine weak learners (i.e., classifiers that perform slightly better than random guessing) to enhance the model's prediction accuracy.

XGBoost builds repetitive DT on the subsample of the training dataset, but at each step, the estimation assigns greater weight to samples that were incorrectly predicted by previous trees. Consequently, every newly created classification tree places more emphasis on the observations that were misclassified by the previous tree. In contrast, RF creates an ensemble of individual trees. Based on the estimation of base tree models, RF computes the results using the majority voting strategy.

For both these ensemble models, it is essential to tune the model parameters that affect the models' performances (i.e., hyperparameters). To tune the hyperparameters, the Grid Search algorithm is utilized. The algorithm uses the cross-validation strategy to split the training data into same-size subsets. Then, it runs the model based on various combinations of the hyperparameters.

Table 2 presents the selected range of each hyperparameter, a brief description of them, and the optimal value found in the Grid Search procedure for each of these models.

Artificial neural network

ANN refers to classifiers whose structures resemble the brain's network structure. While there are numerous approaches to ANN for handling a wide range of input types, this paper explores the feed-forward neural network as a general-purpose algorithm. This algorithm was first introduced by Svozil et al. (1997) and is also known as the Multi-Layer Perception.

This study focuses on the feed-forward neural network as a general-purpose algorithm for classifying commuters' mode choices. A feed-forward neural network is typically trained with a mini-batch gradient descent algorithm (Bottou, 2010). For developing a feed-forward neural

Table 2
Hyperparameter tuning results of ensemble models.

Hyperparameter	Description	Model	Test range	Optimum value
Learning-rate	Step size shrinkage used in each update to prevent overfitting	XGBoost	[0, 1]	0.03
Gamma	Minimum loss reduction required to make a further partition	XGBoost	[0, 5]	1.4
Mini-child-weight	Minimum sum of samples weight needed in a child node	XGBoost	[0, 10]	1
Subsample	Sample ratio of the training dataset used in each iteration	XGBoost	[0, 1]	0.9
Reg-lambda	Regularization terms on weights	XGBoost	[1, 5]	1
Reg-alpha	Regularization terms on weights	XGBoost	[0, 1]	0.1
Max-depth	A limit on how many sequential splits can be applied per branch	XGBoost	[3, 7]	5
Max-depth	A limit on how many sequential splits can be applied per branch	RF	[3, 20]	15
Forest-size	Number of trees in the model	RF	[1,200]	50
Min-samples-split	Minimum number of samples required to split an internal node	RF	[2, 15]	4
Min-samples-leaf	Minimum number of samples required to be at a leaf node	RF	[1, 15]	4
Max-features	The number of features to consider for each tree	RF	[2, 11]	8
Max-samples	The maximum number of samples to consider for each tree	RF	[0.2, 0.9]	0.9

network, unlike the previously mentioned ML algorithms, we use a validation dataset to avoid overfitting (Srivastava et al., 2014; Wang et al., 2020; Ma and Zhang, 2020). This is because the neural network is more prone to overfitting compared to other ML algorithms. Therefore, it is necessary to first split the dataset into three subsets randomly, each containing the same portion of each class. 25 % of the dataset is set aside for testing purposes. To maintain consistency between models, the testing subset is the same as the dataset used to test other models' performance. 60 % of the data is used for training purposes, and 15 % is used for validation.

The model is developed with various numbers of hidden layers ranging from 1 to 5 and different activation functions (i.e., Relu, Sigmoid, softplus, etc.). The best performance is achieved using two hidden layers and one output layer. The most effective activation function for the hidden layers is determined to be Relu. The highest performance is attained by employing 52 neurons distributed across layers. To prevent overfitting, the dropout value of 0.1 demonstrates the best performance, while various dropout values ranging from 0 to 0.5 are tested. As the final hyperparameter, several values for the learning rate (e.g., 0.1, 0.01, 0.001, 0.0001, 0.00001) are tested, which determine how much the network parameters are involved in the training process. The optimal value for the learning rate is determined to be 0.0001.

Resampling

All classification algorithms discussed in previous sections face significant performance challenges when dealing with imbalanced datasets, which are frequently encountered in real-world applications (Ganganwar, 2012). A dataset is considered imbalanced if it contains more samples from one group of target variables than from the rest. This problem results in a small number of training examples for a specific target group while other classes make up the majority. In this scenario, classifiers can be highly accurate for the majority class but very inaccurate for the minority classes. This is due to the influence the larger majority class has on traditional training criteria. On the other hand, most classification algorithms are designed to maximize the fraction of correct predictions to incorrect predictions (i.e., accuracy). These algorithms ignore the differences between types of misclassification errors, specifically assuming that all classification errors cost the same.

The problem of imbalanced datasets is a prevalent issue in the literature on transportation mode choice prediction (Omrani et al., 2015; Wang and Ross, 2018). The imbalanced datasets in the aforementioned studies led to relatively low prediction accuracy for the group of variables that were among the minority group, while their overall model performances were acceptable.

To address this problem, we employ resampling techniques in our modeling. Two resampling approaches considered as solutions to imbalanced datasets are oversampling and undersampling. By using either strategy, the learning system receives the training instances as if they were representative of a well-balanced dataset. As a result, any bias caused by the different percentages of examples per class would be eliminated. Both resampling approaches have significant drawbacks. Under-sampling may lead to the loss of potentially useful data, while oversampling artificially enlarges the dataset, exacerbating the computational burden for the learning algorithm. In this study, we developed two algorithms for each of the resampling techniques to ensure that we achieved the highest performance of our models.

For oversampling, random oversampling (ROS) is developed as the first algorithm, where no new sample is created. Instead, some samples are randomly selected and duplicated to increase the number of samples in minority classes. Additionally, in Synthetic Minority Oversampling (SMOTE) (Chawla et al., 2002), the minority class is oversampled by creating synthetic examples rather than by oversampling with replacement. For undersampling, we first developed the most common algorithm, which is Random UnderSampling (RUS). In this algorithm, samples from the majority classes are randomly deleted from the dataset. Secondly, we used the Tomek Link algorithm (Tomek, 1976), an

enhanced Nearest-Neighbor Rule, to remove the noisy samples from the majority classes.

Performance measurements

Consistent with recent evaluations of ML-calibrated mode-choice models, we therefore report multiple metrics and emphasize robustness checks when interpreting variable effects (Darwish et al., 2024). One method is by presenting the confusion matrix of the model output. The confusion matrix (Stehman, 1997) is an $N \times N$ matrix used for assessing the performance of a classification model, where N is the number of available labels (i.e., transportation modes). The matrix compares the actual labels with those predicted by the model. Thus, it is possible to gain a comprehensive understanding of how a classification model performs and its errors.

Another common measurement is the accuracy index. Accuracy is the number of correctly predicted labels out of the total sample size. The mathematical formulation of accuracy is presented below:

$$\text{Accuracy} = \frac{\sum_j CP_j}{\sum_j n_j} \quad (3)$$

where:

CP_j = the number of correctly classified samples associate with alternative j .

n_j = the total number of samples associated with alternative j .

F1-score is another way to measure the performance of ML classifiers. It is the harmonic mean of precision and recall, which takes into account whether the model is biased toward predicting a specific label or not. The higher the F1-score, the more balanced the model is. The mathematical formulation of F1-score, as well as precision and recall for a single label, is presented below:

$$\begin{aligned} \text{F1-score} &= \frac{2 * \text{precision} * \text{recall}}{\text{precision} + \text{recall}} \\ \text{Recall} &= \frac{\sum_j CP_j}{\sum_j AL_j} \\ \text{Precision} &= \frac{\sum_j CP_j}{\sum_j PL_j} \end{aligned} \quad (4)$$

where:

PL_j = the total number of labels that are predicted as alternative j .

AL_j = the total number of actual labels associated with alternative j .

On top of the mentioned performance measurement indices, the $\rho^2(0)$ and $\rho^2(C)$ indices are calculated for the MNL model. These indices indicate the logit model's overall quality of fit. $\rho^2(0)$ compares the fit of the model with a set of parameters against the model with all parameters set to be 0. In contrast, $\rho^2(C)$ index compares the estimated mode that has a set of parameters K^* with the model that only has constants. The model that only consists of constants is called the market share model. The mathematical formula for calculating $\rho^2(0)$ and $\rho^2(C)$ for the MNL model is presented below.

$$\rho^2(0) = 1 - \frac{L(K^*)}{L(0)} \quad \rho^2(C) = 1 - \frac{L(K^*)}{L(C)} \quad (5)$$

Where:

$L(K^*)$ = log-likelihood for the model with the full vector of estimated parameters.

$K^* L(0)$ = log-likelihood for the model with no parameters, where all parameters are set to 0.

$L(C)$ = log-likelihood for the model with no parameters, where all parameters are set to C .

Results

Descriptive statistic results

A brief description of the sample is provided in Table 3. The distribution of survey respondents is well-balanced among students and university employees (i.e., staff and faculty members). However, we decided to distinguish between staff members and faculty members since the descriptive analysis reveals significant differences in the behavior of these two groups.

Table 4 illustrates the primary transportation mode preferences of different university members. Bus and C-train are determined to be the first and second chosen primary modes for the students, while the majority of both staff and faculty members chose solo occupancy vehicles (SOV) as their primary transportation mode for commuting to and from university campuses. The main difference between staff and faculty members is that staff members use transit more, while faculty members prefer active modes after SOV.

In the mode choice modeling analysis for the university commuters, four modes of transportation are considered: car, bicycle, walk, and transit. Our analysis includes 42.3 % made by car, 43.9 % made by transit, 8.1 % made by walking, and 5.6 % made by biking.

Model estimation results

We considered two levels of comparison for our models. First, we evaluated the accuracy of the models in predicting travel modes for commuters. Since our modeling of travel mode choices was challenged by an unbalanced dataset, we resampled the dataset using the mentioned techniques and re-ran the models to evaluate the effect of resampling in travel choice modeling as well. Second, we compared the feature importance in developed algorithms for evaluating the effects of pre-processing (i.e., how newly generated features and processed features affect the modeling process) and understanding various variables' effects on various models.

Table 3
Sample description.

Variable	Categories	University status share		
		Student N = 430	Staff N = 368	Faculty N = 69
Gender	Male	30.94 %	30.98 %	44.93 %
	Female	69.06 %	69.02 %	55.07 %
Age	<= 29	80.69 %	10.59 %	1.45 %
	> 30	19.31 %	89.41 %	98.55 %
Annual HH income	<= \$30,000	33.25 %	0.54 %	0.0 %
	\$30,000 – \$34,999	5.34 %	1.08 %	0.0 %
	\$35,000 – \$44,999	8.60 %	5.43 %	0.0 %
	\$45,000 – \$54,999	4.65 %	6.52 %	0.0 %
	\$55,000 – \$64,999	5.58 %	6.52 %	1.45 %
	\$65,000 – \$74,999	6.27 %	13.85 %	8.69 %
	\$75,000 – \$84,999	4.41 %	5.97 %	0.0 %
	\$85,000 – \$94,999	14.18 %	20.65 %	14.49 %
	\$95,000 – \$124,999	14.18 %	22.01 %	30.43 %
	>= \$125,000	3.48 %	17.39 %	44.92 %
Car accessibility	Yes	60.93 %	88.58 %	88.40 %
	No	39.07 %	11.42 %	11.60 %
Car ownership	0	20.0 %	8.42 %	8.69 %
	1	26.74 %	39.94 %	40.57 %
2	29.30 %	35.86 %	34.78 %	
	>2	23.95 %	15.78 %	15.96 %
Bike accessibility	Yes	44.88 %	61.41 %	79.71 %
	No	55.12 %	38.59 %	20.29 %
Education Level	Bachelor's student	60.98 %	NA	NA
	Master's student	21.72 %	NA	NA
	Doctoral student	17.30 %	NA	NA

Model prediction performance comparison

An analysis is conducted to determine the highest performing method of predicting the travel mode of university commuters, comparing the DT method, XGBoost, RF, and ANN, considering the base model as MNL. Our selection of these methods was based on their superior predictive performance in the literature and the nature of our dataset.

At the first level of comparison, we compared the actual model prediction performances using the indices explained in Section 3.2.6. The confusion matrix of all models is presented in Fig. 2.

As demonstrated in the above figure, the MNL, DT, and ANN models exhibit significant inaccuracies when attempting to predict lower-frequency modes in the dataset, namely biking and walking. This means that they are prone to bias when predicting modes that are more common in the dataset, that are transit and cars. In contrast, XGBoost showed a more even balance in its mode predictions. Therefore, it may not be appropriate to rely solely on model accuracy when comparing different models, as some may be biased toward predicting specific labels that are more prevalent in the dataset.

The mentioned issue that is caused due to the imbalanced dataset is well seen in the literature. In order to eliminate the effects of an unbalanced dataset, we applied four different resampling methods to our dataset before feeding the data into the models. During the resampling, the test dataset remained unseen and all the procedures were applied to the training dataset in order to avoid data leakage. As mentioned, the resampling methods are categorized into two different groups which either increase the size of the data or reduce it. Fig. 3 compares the results of model predictions on the original dataset as well as on the resampled dataset.

Among the developed models, the MNL model displayed the lowest prediction accuracy when it came to predicting the transportation mode choice of commuters in the original dataset. This is consistent with the literature, which suggests that the MNL model is less successful than ML models in prediction. In contrast, ensemble models outperformed other models in terms of both prediction accuracy and F1-score. Among the ensemble models, the XGBoost model was particularly effective in handling the variability in data and reducing prediction variance across different transportation modes. However, the superior performance of ensemble models is heavily dependent on hyperparameter tuning. Comparing the preliminary XGBoost model results (65.89 % accuracy) with the accuracy of the tuned model (72.8 %) underscores the significance of hyperparameters in predicting the XGBoost model's accuracy. The DT model was tested on the original testing dataset but demonstrated relatively lower performance than tree-ensemble models in terms of both accuracy and F1-score. The ANN model did not perform as well as the ensemble models but exhibited better performance than the MNL and DT models.

When it comes to the resampled dataset, oversampling methods outperformed undersampling methods in terms of accuracy and F1-score. Oversampling had a considerable impact on the overall performance of the prediction models. This means that not only did oversampling improve the prediction accuracy of models, but it also resulted in more balanced predictions. The reason for this is that the original dataset's size is not large enough to tolerate losing some samples during the undersampling process. As a result, increasing the data size using oversampling algorithms is preferred. From the perspective of accuracy and F1-score, resampling had a greater impact on the F1-score, indicating that it improved the models in a way that made them less biased towards specific groups of samples (i.e., labels with high repetition in the dataset). Additionally, it is highlighted that the TOMEK and SMOTE algorithms had better performance compared with RUS and ROS, which randomly decrease and increase the size of the dataset, respectively.

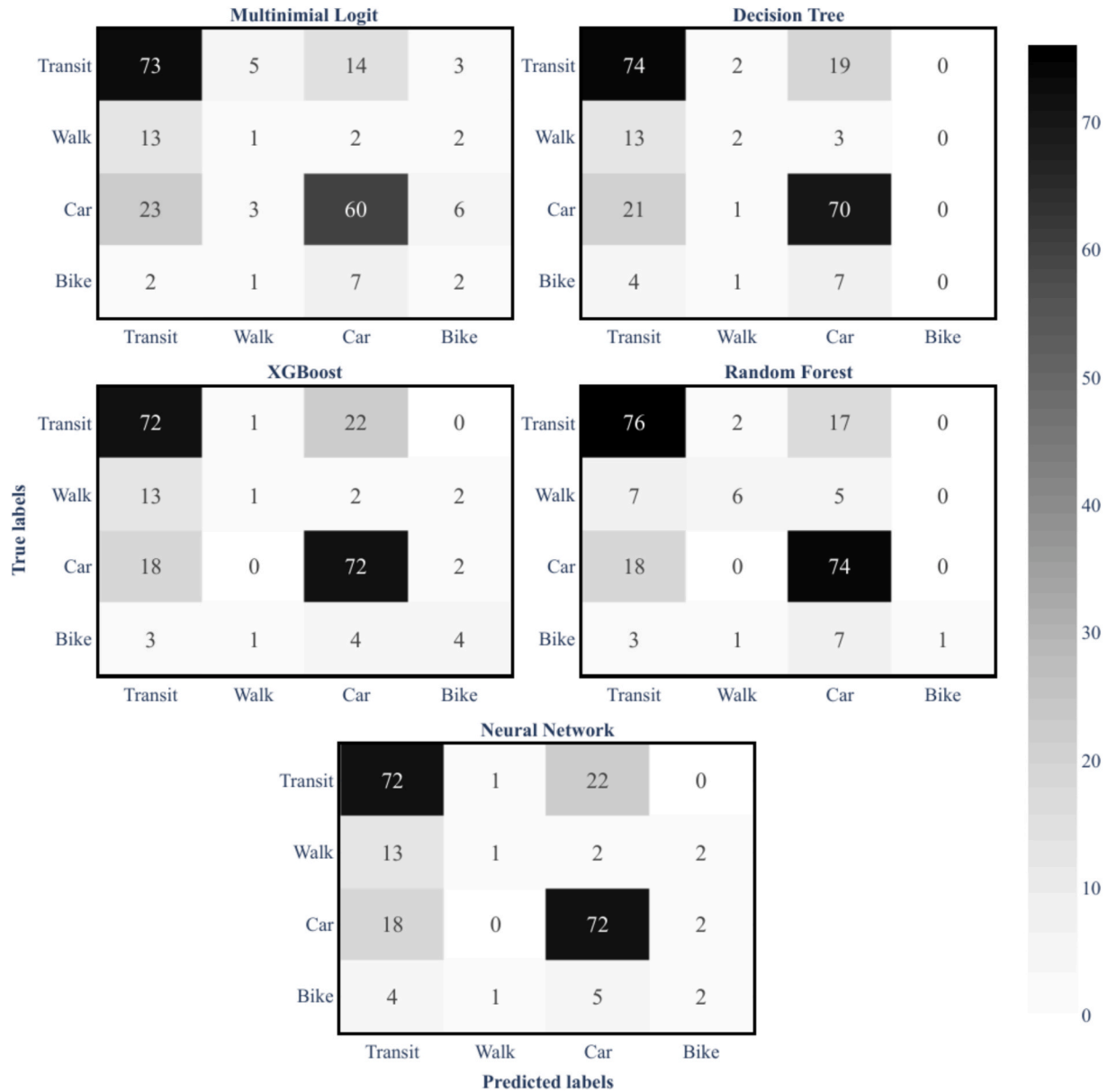
Feature importance comparison

In addition to the models' prediction performance, we estimated the importance of variables in each developed model and compared the

Table 4

Travel mode choice distribution based on university status.

	SOV	Drop off	Carpool	Bus	C-train	Bike	Walk	Others
Student	14.80 %	2.78 %	3.0 %	39.38 %	27.89 %	2.6 %	9.34 %	0.21 %
Staff	39.13 %	1.44 %	2.90 %	7.25 %	13.04 %	18.84 %	14.50 %	2.90 %
Faculty	53.53 %	3.26 %	6.25 %	12.77 %	11.41 %	6.79 %	4.34 %	1.65 %

**Fig. 2.** Confusion matrix.

results together. Since the different models used in this study follow different structures, it is not possible to measure variable importance using the same method for all models.

For the MNL model, which is considered the baseline of our prediction modeling, we evaluated the effect of each variable using the coefficient of each variable along with the t-statistics score and model goodness of fit (i.e., $\rho^2(0), \rho^2(c)$). Overall, the MNL is highly informative in showing the effect of explanatory variables on commuters' mode choice decisions for several reasons. First, the MNL model is able to investigate the effect of each explanatory variable on each alternative utility function separately. Moreover, the MNL reveals whether each

explanatory variable positively or negatively affects the selection of each alternative. These advantages are insightful for professionals and policymakers for future planning. Table 5 represents and compares these values for different variables used in the model.

The estimated MNL results in $\rho^2(0)$ being equal to 0.33 and $\rho^2(c)$ being equal to 0.38 indicating that individuals' preferences variations are well explained in the model. Moreover, the estimated coefficients of the MNL model have consistent signs with the expectations, and most of them have a high level of significance.

Among the features listed in the aforementioned table, travel distance and transit stop/station density are generated and added to the

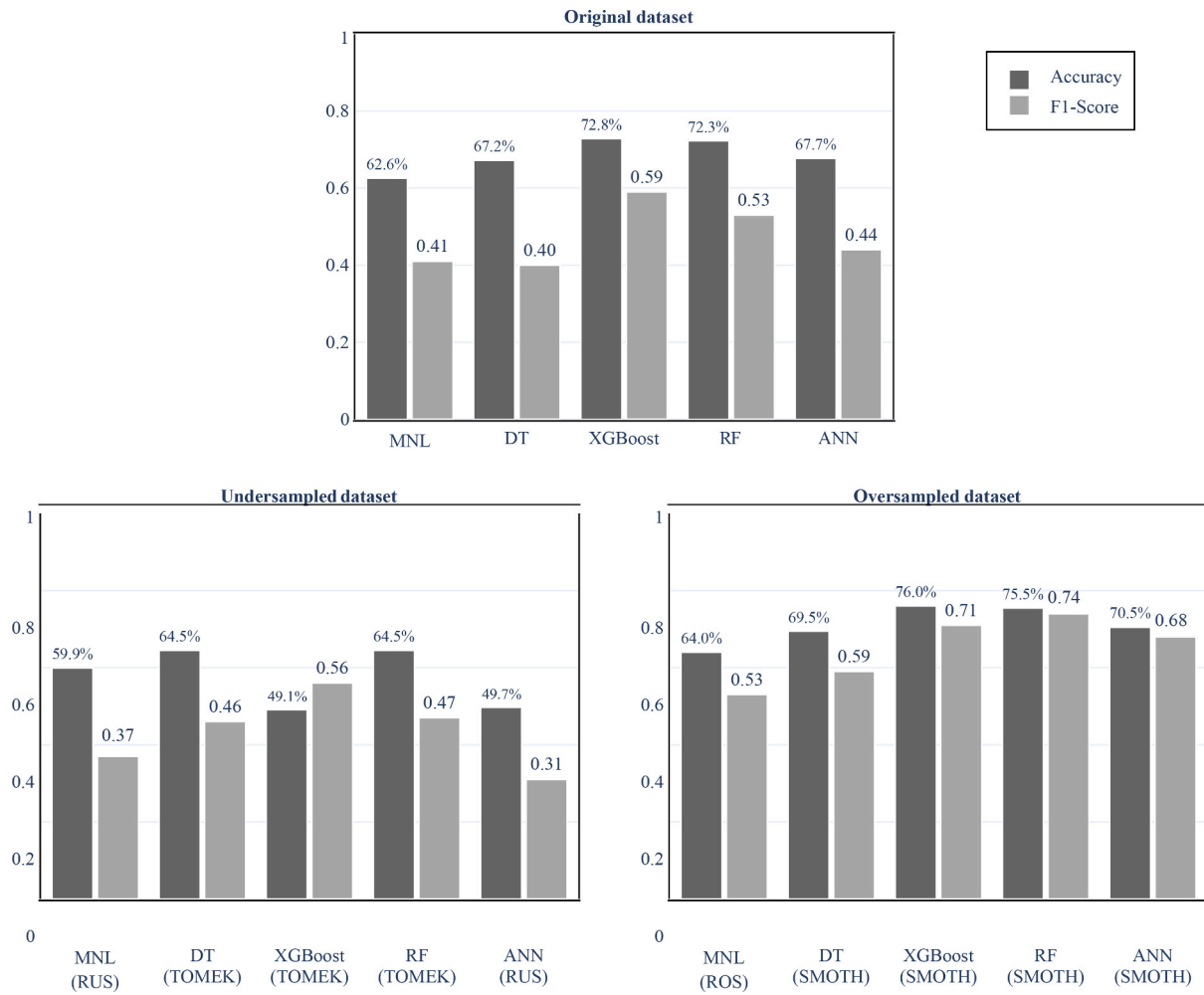


Fig. 3. Predictive models' results.

Table 5
Multinomial model estimation results.

Variable	Transit	t-ratio	Walk	t-ratio	Car	t-ratio	Bike	t-ratio
Travel-Dist (β_1)	-0.06	-0.50	-3.74***	-6.38	0.30**	2.12	-0.03	-0.23
Travel-Freq (β_2)	0.21*	1.88	0.34*	1.69	-0.11	-0.95	0.091	0.77
Season_Effect (β_3)	0.1	0.89	0.26	1.44	-0.26**	-2.13	-0.23**	-1.97
Transit-Pass (β_4)	0.58**	2.44	-0.29	0.88	-0.02	-0.07	-0.20	-0.93
Edu-Lvl (β_5)	0.23	1.11	0.03	0.11	-0.28	-1.45	0.62***	2.72
Gender (β_6)	0.05	0.46	-0.03	-0.16	0.09	-1.48	-0.17	-1.48
Age (β_7)	-0.56***	-3.4	-0.38	-1.39	0.28*	1.67	-0.04	-0.26
Car-Own (β_8)	-1.15***	-6.99	-0.8***	-3.16	0.15	0.93	-0.39***	-2.57
Size (β_9)	0.46***	3.39	0.64***	2.7	-0.08	-0.50	-0.03	-0.24
Income (β_{10})	-0.32**	-2.31	-0.38	-1.46	0.05	0.28	-0.1	-0.68
Bus-Density (β_{11})	0.16	1.0 4	NA	NA	NA	NA	NA	NA
LRT-Density (β_{12})	0.05	0.39	NA	NA	NA	NA	NA	NA

$\rho^2(0) = 0.33$.

$\rho^2(c) = 0.38$.

*P-value < 0.10, **P-value < 0.05, ***P-value < 0.01.

dataset during the feature generation process. Despite these features being derived from large postal code areas and shared among extensive groups of respondents, travel distance has been shown to have the highest impact on choosing car and walk as primary modes of transportation. Transit density-related features also exhibit a positive influence on selecting transit as the primary mode. These results demonstrate that generating travel-related features, even with limited available resources, can enhance the performance of mode predictions.

For the transit mode specifically, the importance weight of features is

evenly distributed among the available variables, with car ownership having the highest absolute impact. The availability of a transit pass is the second most influential factor in selecting the transit mode. In line with transit pass availability, household income has a significant negative impact on selecting transit as the primary mode. Policymakers can further examine these relationships in order to incentivize commuters to use public transit more frequently.

In walk mode, travel distance is the most significant factor influencing commuters' choice, which can be attributed to the city's weather

conditions and the nature of this mode of transportation. The number of cars available in a household is the second most influential factor, reducing the likelihood of commuters walking to campuses. Similarly, car ownership has a negative and significant impact on both biking and transit modes. This implies that the availability of a car substantially decreases the likelihood of commuters using alternative modes of transportation. Household size and travel frequency are other two significant attributes in selecting walk as the primary mode of transportation. The positive sign of household size can be explained by the negative sign of car ownership, meaning that as the number of people living in a household increases, there would be fewer cars available to use. The positive sign of travel frequency is also in line with the negative sign of travel distance, meaning that commuters who have higher travel frequency tend to reside in residential areas closer to where they work.

Travel frequency is also shown to contribute positively to transit and walk utility (significant for transit at $p < 0.1$), but negatively to car utility. This implies frequent campus commuters are more likely to choose transit or walking, all else equal, a pattern captured thanks to the alternative-specific coefficients in our MNL specification.

In contrast to transit and walk modes, household attributes such as car ownership, household size, and income level have low and insignificant impacts on choosing car as the primary mode of transportation. Instead, travel distance, season effect, and age are significant factors for choosing car as the primary mode of transportation. Travel distance and age positively affect car selection, while they negatively affect the choice of other modes of transportation. Season effect (Yes = 1 if the individual's commuting mode changes from summer to winter, 0 otherwise) was included in the MNL model with separate coefficients for each mode. This allows seasonal travel behavior changes to affect each transportation mode's utility differently. For instance, seasonal change has a significant negative effect on choosing a car, indicating that respondents who adjust their commuting habits between summer and winter are less likely to drive. This finding is intuitive, as harsh winter conditions in central Alberta can reduce the attractiveness of driving. Central Alberta receives on average 55 snowy days every year (Environment Canada, 2019). The snow may remain on roads for longer, especially during fall and winter when the temperature is below zero. This will lead to people being concerned about the poor performance of cars under adverse weather conditions, such as snow and rain.

In the ML models, we evaluated the variable importance in predicting the results for each model. For tree-based models (i.e., DT, XGBoost, and RF), we assessed the importance of variables using impurity drop. Meanwhile, due to the differences in the structures of these models, this study focuses on permutation importance for the neural network model. The top 10 variables considered to be the most important features in training the models are compared in Table 6.

The findings from the analysis of various ML models provide valuable insights into the factors that influence commuters' choice of transportation mode. A deeper understanding of these factors can help policymakers and planners develop targeted strategies to encourage the

use of more sustainable modes of transportation.

In tree-based models, the most important variables are accessibility to a car and access to a transit pass. These boolean variables play a significant role in determining the choice of transportation mode, highlighting the importance of boolean variables in tree-structured based models. Ensemble models mitigate the bias toward specific variables, which, in turn, further improves model performance. A comparison of the XGBoost and RF models reveals that the RF model is more successful in removing bias toward available attributes, consistent with Fig. 3, which shows the highest F1-score among developed models for the RF model.

In contrast, the neural network model places more emphasis on continuous variables, such as the number of cars owned by the household, rather than boolean attributes such as car accessibility. This difference in variable importance highlights the varying ways that ML models process and prioritize information when making predictions.

Travel-related information, including travel distance, travel duration, and transit stop/station density, proves to be highly influential across all ML models. This finding underscores the value of generating new variables that capture travel-related aspects to improve the accuracy of predicting travel mode choices. By understanding the significance of these variables, policymakers can identify potential areas of intervention and design data-driven solutions to promote sustainable transportation choices among commuters.

In summary, the paper's contributions are threefold. First, we quantify the performance uplift from tuned ensembles: XGBoost and Random Forest reach 72.8 % and 72.3 % accuracy versus 62.6 % for MNL; with oversampling, accuracy increases to 76.0 % and 75.5 %, and F1 rises to 0.71 and 0.74 compared with 0.41 for MNL. Second, we show cross-model agreement on the key drivers of choice with explicit magnitudes: in the decision tree, car access accounts for 36.1 % of total importance, about 1.5 times transit pass at 23.3 % and more than three times transit distance at 9.9 %; XGBoost ranks transit pass at 17.1 % and car access at 15.2 %; Random Forest places car access and transit pass at 9.1 % and 8.8 %. The MNL yields expected signs and solid fit with $\rho_2(0) = 0.33$ and $\rho_2(c) = 0.38$. Third, we demonstrate stability and actionability: effect directions are consistent across model families, persist under hyperparameter tuning and class-balance adjustments, and align with the out-of-sample gains. Together, these results isolate the dominant signals, quantify their strength relative to convenience variables, and set up the conclusions that follow.

Conclusion

This study examined the data from the University commuting habits survey conducted in March 2020, with a twofold purpose: (1) developing diverse predictive models to predict university commuters' transportation mode choices, thereby evaluating the predictive performance of ML models in comparison to the traditional MNL model, and (2) investigating the relative importance of explanatory features to

Table 6
Feature importance of machine learning models.

Rank	Decision tree		XGBoost		Random forest		Neural network	
	Variable	Score	Variable	Score	Variable	Score	Variable	Score
1	Car-Acc	36.1 %	Transit-Pass	17.1 %	Car-Acc	9.1 %	Transit-Pass	NA
2	Transit-Pass	23.3 %	Car-Acc	15.2 %	Transit-Pass	8.8 %	Car-Own	NA
3	Transit-Dist	9.9 %	Uni-Status	6.3 %	Income	8.6 %	Walk-Dist	NA
4	Income	9.1 %	Drive-Dist	6.2 %	Uni-Status	6.4 %	Travel-Freq	NA
5	Bike-Dist	6.9 %	Bus-Density	4.3 %	Car-Own	6.3 %	Age	NA
6	Car-Own	3.2 %	Drive-Dur	3.1 %	Transit-Dist	5.1 %	Edu-level	NA
7	Bus-Density	3.1 %	Transit-Dist	3.1 %	Transit-Dur	4.3 %	Income	NA
8	Travel-Freq	3.0 %	Bike-Acc	3.0 %	Age	4.2 %	Car-Acc	NA
9	Bike-Dur	2.8 %	Edu-Level	2.9 %	Dirve-Dur	3.7 %	Bus-Stops	NA
10	Transit-Dur	2.6 %	Car-Own	2.9 %	Drive-Dist	3.7 %	Gender	NA

*Sum of the score might not be equal to 100% since only the top 10 variables are presented.

better understand the influential variables on university commuters' mode choice behavior in different models.

To ensure result maturity and guard against overfitting or modeling artifacts, we implemented a multi-layered validation pipeline. First, we used an ethics-approved, university-wide survey (979 responses; 867 retained after pre-processing) and engineered travel-time/distance and transit-access variables by combining Google Distance Matrix estimates from multiple points per postal code with GIS-based bus/LRT stop densities. We encoded categorical features appropriately (ordered vs. one-hot) and specified alternative-specific terms in the MNL; 47 candidate specifications were tested before selecting the final utility structure. For tree-based models (DT, RF, XGBoost) we pruned/tuned with stratified 10-fold cross-validation and grid search; for the ANN we used a fixed 60/15/25 train/validation/test split with dropout and a low learning rate. Because mode shares were imbalanced, we applied both under and over-sampling (RUS, Tomek, ROS, SMOTE) strictly within the training folds to avoid leakage. Model performance was evaluated with accuracy and F1-score and visualized via confusion matrices, and we compared feature importance across algorithms against MNL signs and significance to check the stability of substantive conclusions. Collectively, these steps, i.e. data engineering, alternative model families, cross-validated tuning, class-imbalance handling, and multi-metric reporting, were undertaken to substantiate the robustness and maturity of the results.

The results of this study show that some travel-related features such as travel time and travel distance are significant in model development, especially in mode choices of car and walk. On the choice of public transit, it was observed that car ownership and availability of transit passes are the most influential factors in the context of university commuters. As such, policymakers can further examine these relationships in order to incentivize commuters to promote public transit. It was also observed that, in contrast to transit and walk modes, household attributes such as car ownership, household size, and income level have low and insignificant impacts on choosing car as the primary mode of transportation.

These findings are particularly important and unique for the case of university commuters as the commuters exhibit different mobility behaviors compared to the general population. Previous studies investigating university commuters are primarily limited to descriptive analysis and did not develop prediction models to investigate the factors affecting university commuters' transportation mode choice decisions. The main objective of this survey was to understand the main factors that prevent commuters from choosing more sustainable transportation modes and change policies to shift further toward green transportation. As such, understanding the explanatory features for mode choice decisions was critical. Thereby, we employed both ML models and traditional MNL models to compare the predictability and to understand important features for mode choices.

The machine learning models yield additional insights into commuter behavior that complement the MNL findings. Notably, binary access factors emerge as dominant: for example, the decision tree model shows that having access to a car accounts for a large share of its splits (36 % of total importance), far exceeding any other single factor. This underscores that simply whether an individual has a car or a transit pass can decisively shape their travel mode. In practice, this suggests that university policies like restricting campus parking (to reduce car availability) or providing universal transit passes could significantly influence mode choices. Additionally, the ML models uniformly highlight travel convenience variables, such as commute distance, travel time, and transit-stop density as key predictors. These factors' high importance signals that improvements in transit service coverage, travel time, or distance (for instance, through satellite campuses or better biking infrastructure) may substantially encourage shifts away from driving.

In addition to confirming these known determinants, the ML methods also revealed behavioral patterns that the linear MNL structure could not capture. Specifically, ML models identified non-linear

thresholds such as sharp declines in walking and biking beyond certain commute distances, and interaction effects between car access and transit accessibility (e.g., the effect of car ownership being strongest when transit stop density is low). The neural network model further highlighted the importance of car ownership intensity, showing that additional vehicles per household yield diminishing increases in driving likelihood. These non-linear and high-dimensional patterns, automatically captured by ML, uncover more realistic behavioral responses and help identify where policies would have the greatest marginal effect.

Importantly, we observe that different ML algorithms prioritize information in complementary ways. Tree-based models focus on threshold effects (e.g., whether a commuter has a car or not), whereas the neural network gives more weight to gradual effects (e.g., the number of vehicles in the household). This suggests that beyond the binary presence of a car, the extent of car access also influences mode choice, a nuance that aligns with intuitive behavior patterns. By recognizing these non-linear relationships, ML results provide more precise targets for policy. For instance, identifying critical commute distance ranges or car-access levels where interventions such as parking restrictions, universal transit passes, or last-mile shuttle enhancements would be most effective. Such insights, captured through non-linear ML techniques, enrich our understanding of mode choice determinants and provide actionable guidance: policies targeting both the existence and the degree of private vehicle access, as well as the overall convenience of alternative modes, can be effective in steering university commuters toward more sustainable transportation options.

We identify several important findings regarding the strengths and limitations of MNL and ML models in this study. First, our analysis of feature importance in different predictive models highlighted that the MNL model delivers a deeper understanding of the relationships between explanatory variables compared to ML models. This is primarily because the MNL model allows for the examination of each transportation mode individually, offering a more detailed perspective on how various factors contribute to commuters' mode choice decisions. Second, the MNL model is capable of discerning the statistical significance of studied variables, shedding light on the relative importance of each factor in shaping commuters' preferences and behaviors. ML models, however, were found to prioritize variables differently when generating predictions. As such, depending on the desired perspective of the study, either MNL or ML models can be utilized for the purpose of the study. Third, the availability of explanatory data can significantly influence the performance of ML models, as their inherent structure and learning algorithms determine how they prioritize various factors when making predictions. Thus, understanding the data and selecting an appropriate model that aligns with the characteristics of the explanatory variables is crucial for achieving accurate and reliable predictions. In summary, both MNL and ML models can be complementary and enhance the insights derived from the data analysis.

This study contributes to interdisciplinary transportation research by comparing behavioral choice modeling with data-driven machine learning and policy analysis in a university commuting context. The design integrates social-science constructs (attitudes, constraints, and built-environment access) with engineered travel metrics and compares model families to triangulate effects across complementary lenses. The case-study orientation is practice-relevant (transportation demand management and emissions mitigation), and the comparative modeling provides transferable evidence for analysts and practitioners. By situating a local dataset within broader questions of mode choice, evaluation, and policy, the methods and checks reported here are intended to strengthen inference and make the results actionable beyond the immediate setting.

Our analysis indicates that comparing theory-driven choice models with modern machine-learning classifiers can yield complementary insights for campus commuting policy. The consistency we observe across model families on the roles of access, travel time, and built-environment context suggests that the findings are not artifacts of a single

specification. Framing the study as a decision-support tool highlights practical levers for transportation demand management, while the comparative approach clarifies where additional data (e.g., explicit monetary costs) would most improve predictive performance and policy sensitivity. Taken together, the evidence offers a grounded basis for near-term interventions and a pathway for scaling similar evaluations in other institutional settings.

Beyond the comparative analysis of MNL and ML models, this study also yields several important recommendations for policy implications. One of our most robust findings is that having a transit pass (e.g., a university UPass) strongly encourages transit use. Those with transit passes were far more likely to choose transit over driving. This suggests that continuing and expanding subsidized or mandatory transit pass programs can be highly effective. A policy implication is that the university should ensure all students (and possibly staff/faculty) have access to an affordable transit pass. Prior research similarly noted that subsidized public transport passes are among the most critical factors in shifting commuters to transit. Our analysis reinforces that maintaining a universal transit pass program (or extending it to employee groups) would likely reduce single-occupancy car trips to campus.

Our results show that car availability (e.g., car ownership or having a car accessible) is a major determinant of driving. Results show that higher car ownership/access leads to higher likelihood of driving and lower use of alternative modes. While the university cannot directly control how many cars a household owns, it can implement policies that indirectly reduce convenient car access for commuters. For example, the university could raise parking fees or limit parking permits for certain groups. The literature indicates that higher parking costs and restricted parking availability discourage driving to campus. Based on our findings, policies such as capping the number of parking permits (especially for students living on or near campus), increasing parking charges, or providing incentives for those who forgo bringing a car to campus could reduce drive-alone commuting. Another approach is to designate “car-free” days or zones on campus or prioritize carpool parking permits to encourage shared rides. These strategies target the ease of driving and can make alternative modes more competitive.

We found that the density of transit stops/stations near one’s home significantly influences transit use (higher transit stop density around home correlates with a greater likelihood of choosing transit). This suggests that improving transit accessibility for areas where students and staff live can boost transit ridership. Policy-wise, the university could work with the city transit agency to ensure high-frequency transit routes serve the campus and that there are adequate bus stops or LRT stations near campus and in student-populated neighborhoods. Investing in convenient, reliable campus shuttle services or last-mile connections from transit hubs to campus could also encourage those who might otherwise drive.

Our analysis indicates that travel distance is a key barrier for walking and cycling. The study shows that longer distances sharply reduce the likelihood of walking or biking to campus. Additionally, seasonal effects (winter weather) make bike use less appealing for many. To overcome some of these barriers, the university could implement policies such as improving on-campus housing availability or incentives for living close to campus (which shortens commute distances). More directly, the university can enhance active transport facilities: build safe bike lanes connecting campus with nearby districts, provide secure bicycle parking, lockers and showers for cyclists, and ensure sidewalks and bike paths are well-maintained and cleared of snow in winter. By improving the comfort and safety of walking and cycling (even in less favorable seasons), these modes become more viable. For example, a policy to increase winter maintenance of pedestrian paths and cycling routes would address weather-related deterrents, aligning with our finding that seasonal concerns affect mode choice. Universities like ours could also promote initiatives like campus bike-share or e-scooter programs to facilitate short-distance travel without cars.

Moreover, we show some differences among demographic groups (e.

g., younger commuters or students tend to use transit more, whereas older faculty/staff have higher car usage rates, partially due to higher income and more parking permits). A potential policy is to target new students and employees with travel behavior programs. For instance, an orientation program highlighting transit options, car-share services, or a “try transit/bike week” campaign. Additionally, since income and convenience influence mode choice, the university might consider incentive programs such as discounted transit for employees or reward schemes for those who commute sustainably (for example, a certain number of transit or bike commutes earns a benefit). These tailored strategies can nudge those who currently drive (especially when they have easy car access) to consider alternative modes.

CRediT authorship contribution statement

Hassan Kamkar: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Saeid Saidi:** Writing – review & editing, Writing – original draft, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Conceptualization. **Mohammad Ansari Esfeh:** Writing – review & editing, Writing – original draft, Methodology, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

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